

Measuring Learning in Informative Processes

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Working draft

Abstract

Observed knowledge gain can be a poor indicator of learning from broadly informative processes. When the knowledgeable learn more, easy questionnaires give those who learn the most the least room to show it. We derive conditions under which post-process knowledge correlates more strongly with true gain than observed gain does. A general monotone measurement model also guarantees nonnegative covariance between post-process knowledge and true gain under stated learning and independence assumptions. In 1,212 proportional-learning simulations whose observable summaries fall within the ranges of 21 Deliberative Polls, post-process knowledge outperforms observed gain in 100% of cases; their median correlations with true gain are .49 and .01, respectively. Scoring and baseline adjustment also change associations with education and attitude change in the polls. Observed gain performs better in the catch-up simulations. In most polls, when questions are randomly divided into two sets, observed gain on one set correlates more strongly with observed gain on the other than the follow-up score does.

1 Introduction

Many communication processes are broadly informative, exposing their participants to some goodly quantity and wide variety of information on at least one sizable topic, like policy or electoral choices. The category includes both natural processes like election campaigns and ad hoc but naturalistic ones like Deliberative Polls. The information exposure need not be the point. Often, it is not. In election campaigns, it is incidental, a by-product of persuasion or mobilization. In Deliberative Polls and other deliberative fora, it is intrinsic but instrumental—not an end in itself but an ingredient in more fully considered, better founded opinions. But in whatever fashion, for whatever reason, the exposure must occur, and at least some participants must learn something from it, or the process would be only degenerately “informative.”

Almost always, we are interested in this learning—sometimes in the learning of this or that particular bit, affecting this or that particular attitude (Gilens 2001), but more often in the amount of topic-wide learning, affecting a wider range of attitudes, as well as vote intentions, levels of engagement, and the like (Bartels 1996; Delli Carpini and Keeter 1996; Althaus 2003). In the aggregate, this is also normatively important, speaking to the educative value of election campaigns and the deliberative quality of deliberative fora.

But how best to measure it? The obvious choice is observed knowledge gain: the increase in the proportion of knowledge items answered correctly. For aggregate description, *faute de mieux*, this may serve. But individual-level analyses—of learning as a function of causally prior variables or of causally subsequent ones as a function of learning—are another story. There, observed gain can be a weak indicator of true gain, and post-process knowledge can be much better. When the knowledgeable learn more, those with the most to gain may already be near the questionnaire’s

ceiling. We derive conditions under which post-process knowledge has the higher correlation with true gain and show that its positive association extends to general monotone measurement functions. In the retained proportional-learning simulations, its advantage over observed gain holds in 100% of cases. Scoring and baseline adjustment also change who appears to learn and how learning relates to attitude change in the polls.

2 Observed and True Knowledge

The knowledge we are used to thinking about is “observed.” It is, by definition, all we ever see. Conventionally, the i th person’s observed knowledge at time t —call it x_{it} —is the proportion of the questionnaire’s knowledge items the person answers correctly. We are much less used to thinking about “true knowledge”—what observed knowledge is trying to observe. Its depths can be left to philosophy and neuroscience. Nearer the surface, we may define the i th person’s true knowledge at time t —call it X_{it} —as the proportion of all knowable, relevant items he or she knows at time t . Observed knowledge gain is $\Delta x_i = x_{i2} - x_{i1}$ and true knowledge gain $\Delta X_i = X_{i2} - X_{i1}$.

A broad domain of policy knowledge includes far more than a questionnaire asks. A person may answer every question correctly while knowing only a small share of the domain’s facts. We model this disparity with low, right-skewed true knowledge and questionnaires concentrated on easy items.

Let Q denote the questionnaire’s items, U the domain of relevant items, $K_{it} \subseteq U$ the items the person knows, and C_{it} the questionnaire items answered correctly. Then $x_{it} = |C_{it}|/|Q|$ and $X_{it} = |K_{it}|/|U|$. The proportions require a common unit of knowledge and $Q \subseteq U$; compound questions must be mapped to those units. The domain and counting rule fix the denominator.

Observed knowledge differs from true knowledge in both its domain and what it tallies: Q , just the questionnaire’s items, versus U , the relevant domain; and correct responses versus items known. Correspondingly, the measurement error is of two sorts. Write $X_{it}^Q = |K_{it} \cap Q|/|Q|$. The response-production error lies in the responses to given items: lucky guessing can produce a correct answer without knowledge, and reticence can produce a “don’t know” response despite knowledge. The item-sampling error lies in Q ’s subsetting of U . At the index level,

$$x_{it} - X_{it} = \underbrace{x_{it} - X_{it}^Q}_{\text{response-production error}} + \underbrace{X_{it}^Q - X_{it}}_{\text{item-sampling error}} .$$

A correction for guessing addresses the first term. It does not, by itself, address the second.

The distinction matters for change as well as levels. Assuming away guessing, reticence, and other response error, observed gain is valid if we intend to constrain our inference to the items asked. If $Q = U$, there is no item-sampling error. If Q is a simple random sample from a fixed, finite U , the questionnaire gain is unbiased for domain-wide gain over repeated item samples. A short random questionnaire can nevertheless rank individual learners poorly because of sampling variation. A longer questionnaire drawn repeatedly from the same unrepresentative part of a domain can be precise about that part and misleading about the rest.

The domain also determines what a ceiling means. In an experiment with only a handful of facts to learn, a perfect score may represent exhaustion of the learning domain. In a broadly informative process, a person can learn much more while continuing to answer every questionnaire item correctly. Even among people whose observed scores move from zero to one, the amount learned outside the questionnaire can differ. A questionnaire ceiling limits what can be shown; it need not limit what can be learned.

The initial models use the following assumptions, setting aside guessing and other response error:

- L1. At each positive initial knowledge level, participants learn on average: $E(X_{i2}|X_{i1}) > X_{i1}$ for $X_{i1} > 0$.
- L2. The more knowledgeable learn more: $E(\Delta X_i|X_{i1})$ increases with X_{i1} . More new information sticks when there is more previously cognized information for it to stick to (Recht and Leslie 1988; Eckhardt et al. 1991; Cooke et al. 1993; Hambrick 2003).
- L3. Those who know nothing learn nothing.
- M1. Those who know nothing show no knowledge, and those who know the most show as much as the measure permits.
- M2. The more knowledgeable appear to know more, but the less so, the more they know: $E(x_{it}|X_{it})$ increases with X_{it} but flattens as it approaches its ceiling of 1.
- M3. At positive knowledge levels in the modeled range, people appear to know more than they do: $E(x_{it}|X_{it}) > X_{it}$.
- M4. The more knowledgeable appear to learn more, then, beyond a point, to learn less. Those who know nothing actually learn nothing; those who know the most, despite learning the most on average, cannot show it, having started at or near the ceiling.

Three phenomena motivate this regime.

Item-sampling bias. In this regime, the questionnaire’s knowledge items are a nonrandom draw from the domain of relevant, knowable items, concentrated on its easier parts. Test designers may favor questions that anyone paying a modicum of attention to the news should know, since a battery of expert questions would leave most respondents near zero except by lucky guessing. This can let relatively knowledgeable respondents ace the questionnaire while knowing only a small fraction of the specified domain.

Ceiling effects. The participants who already know a lot, having a relatively high X_{i1} and thus an absolutely high x_{i1} , cannot show much observed knowledge gain. If the domain extends well beyond the questionnaire, there can still be ample room for true knowledge to increase.

The rich get richer. Knowing more about something makes it easier to learn still more (Dooling and Lachman 1971; Bransford and Johnson 1972; Recht and Leslie 1988; Eckhardt et al. 1991; Cooke et al. 1993; Hambrick 2003). Learning is easier when the material falls effortlessly into context. Hence the participants who cannot show much observed knowledge gain are precisely those whose expected true knowledge gain should be greatest.

These considerations yield two second-blush intuitions. First, ceiling effects and the rich getting richer should depress the correlation between Δx_i and ΔX_i . Second, x_{i2} may be a better indicator. Under the assumed learning process, those who emerge knowing a lot may have gained a lot—either observably, if they initially knew only a little, or unobservably, if they initially knew a lot.

3 A Linear Model

Let

$$x_{i1} = a_1 + b_1 X_{i1} + u_{i1}, \tag{1}$$

$$x_{i2} = a_2 + b_2 X_{i2} + u_{i2}, \tag{2}$$

$$X_{i2} = \alpha + \beta X_{i1} + v_i, \tag{3}$$

implying

$$\Delta x_i = (a_2 - a_1) + b_2\alpha + (b_2\beta - b_1)X_{i1} + (u_{i2} - u_{i1}) + b_2v_i, \quad (4)$$

$$\Delta X_i = \alpha + (\beta - 1)X_{i1} + v_i. \quad (5)$$

For the conditional-mean interpretations, assume $E(v_i | X_{i1}) = 0$ and $E(u_{it} | X_{i1}, v_i) = 0$, with u_{i1} and u_{i2} uncorrelated. These assumptions imply the zero covariances used below; zero covariances alone would suffice for the covariance calculations. The variances $\sigma_{X_1}^2$, $\sigma_{u_1}^2$, $\sigma_{u_2}^2$ and σ_v^2 are all strictly positive: there is some measurement error at each wave, true initial knowledge does not perfectly predict true later knowledge, and true initial knowledge varies. The linear equations are local approximations, not a globally bounded response model. Imposing L1 additionally requires $\alpha + (\beta - 1)X_{i1} > 0$ over the positive initial-knowledge support; the covariance results below do not depend on α .

A second layer of assumptions expresses the three phenomena. *Item-sampling bias*. Those who know nothing, and do not guess, score 0, so $a_t = 0$ and $b_t = E(x_{it}|X_{it} = X_{Mt})/X_{Mt}$, where X_{Mt} is the highest true knowledge in the population. Since the questionnaire lets $E(x_{it}|X_{it} = X_{Mt})$ approach 1 while X_{Mt} is near 0,

$$b_1, b_2 > 1. \quad (6)$$

The rich get richer. A given difference in true initial knowledge results in a bigger difference in true later knowledge:

$$\beta > 1. \quad (7)$$

Ceiling effects. The more one knows, the less one can appear to gain: X_{i1} 's coefficient in (4) is negative,

$$b_1 > b_2\beta. \quad (8)$$

The correlations of observed gain and of post-process knowledge with true gain are $\rho_{\Delta x \Delta X} = \sigma_{\Delta x \Delta X} / (\sigma_{\Delta x} \sigma_{\Delta X})$ and $\rho_{x_2 \Delta X} = \sigma_{x_2 \Delta X} / (\sigma_{x_2} \sigma_{\Delta X})$, where single-subscripted, squared σ 's are variances and double-subscripted ones covariances. From (1)–(5),

$$\sigma_{\Delta x \Delta X} = (\beta - 1)(b_2\beta - b_1)\sigma_{X_1}^2 + b_2\sigma_v^2, \quad (9)$$

$$\sigma_{x_2 \Delta X} = (\beta - 1)b_2\beta\sigma_{X_1}^2 + b_2\sigma_v^2, \quad (10)$$

$$\sigma_{\Delta x}^2 = (b_2\beta - b_1)^2\sigma_{X_1}^2 + \sigma_{u_1}^2 + \sigma_{u_2}^2 + b_2^2\sigma_v^2, \quad (11)$$

$$\sigma_{x_2}^2 = b_2^2\beta^2\sigma_{X_1}^2 + b_2^2\sigma_v^2 + \sigma_{u_2}^2. \quad (12)$$

Three results follow (proofs in Appendix A).

Result 1. $\rho_{\Delta x \Delta X}$ can be negative. Specifically,

- Condition 1a: $\rho_{\Delta x \Delta X} < 0$ if and only if $(\beta - 1)(b_1 - b_2\beta)\sigma_{X_1}^2 - b_2\sigma_v^2 > 0$.

Result 2. $\rho_{x_2 \Delta X}$ is always positive.

Result 3. $\rho_{\Delta x \Delta X}$ can be less than $\rho_{x_2 \Delta X}$. Each of the following is sufficient:

- Condition 3a: $(b_1 - 2b_2\beta)b_1\sigma_{X_1}^2 + \sigma_{u_1}^2 > 0$;
- Condition 3b: $b_1 \geq 2b_2\beta$;
- Condition 3c: Condition 1a;
- Condition 3d: $\gamma_{21} < .5$, where $\gamma_{21} = \sigma_{x_1 x_2} / \sigma_{x_1}^2$ is the slope of the population regression of x_{i2} on x_{i1} .

Three points need noting. First, $\rho_{\Delta x \Delta X}$ need not be negative to be depressed: as the parameters approach a configuration under which it is negative, observed gain is enfeebled, positively but weakly correlated with true gain. Second, none of Conditions 3a–3d is necessary. Third, Condition 3b implies 3a, but Conditions 3a and 3c do not imply one another; neither is uniformly easier to satisfy. The slope in Condition 3d can be estimated from observed scores: in every one of the 21 Deliberative Polls we examine below, the observed correlation of x_{i1} and x_{i2} is between .11 and .72, and the regression slope of x_{i2} on x_{i1} is below .5 in 10 of them. This observable comparison does not verify the model’s assumptions about true learning or measurement error.

Guessing. Let people who do not know an item answer it correctly with probability κr_i , where r_i is the person’s propensity to guess, the same at both waves and independent of the other variables, and κ the chance of a lucky guess. Averaging over guessing outcomes gives $y_{it} = x_{it}(1 - \kappa r_i) + \kappa r_i$. Then $\sigma_{\Delta y \Delta X} = (1 - \kappa E(r_i)) \sigma_{\Delta x \Delta X}$ and $\sigma_{y_2 \Delta X} = (1 - \kappa E(r_i)) \sigma_{x_2 \Delta X}$: guessing scales both covariances by the same nonnegative factor. Realized guessing adds conditionally mean-zero variation and therefore preserves these covariance expressions. Results 1 and 2 retain their strict signs when the factor is positive; guessing correctly with probability one makes the observed score constant.

4 A Nonlinear Model

The knowledge proportions x_{i1} and x_{i2} are bounded by 0 and 1 and often approach the ceiling; their difference lies between -1 and 1. Their consequently decreasing slopes on true knowledge (M2’s ceiling) are a major part of the story, which the linear model captures only through $b_1 > b_2\beta$. A quadratic measurement equation represents the flattening near the ceiling over a restricted range; it does not itself impose a global bound on realized scores. Let

$$X_{i2} = \beta X_{i1}, \quad (13)$$

$$x_{it} = bX_{it} + cX_{it}^2 + u_{it}, \quad t = 1, 2, \quad (14)$$

with $\beta > 1$, $b > 0$ and $c < 0$. Assume $E(u_{it} | X_{i1}) = 0$ and uncorrelated errors across waves, so that the errors are also uncorrelated with X_{i1}^2 . Then

$$\Delta X_i = (\beta - 1)X_{i1}, \quad (15)$$

$$\Delta x_i = (\beta - 1)bX_{i1} + (\beta^2 - 1)cX_{i1}^2 + u_{i2} - u_{i1}. \quad (16)$$

M2 requires observed knowledge to rise with true knowledge over the range of true knowledge at both waves: $b + 2cX_{it} > 0$, or $X_{M2} = \beta X_{M1} < -b/(2c)$; equality permits a zero derivative at the endpoint. $E(\Delta x_i | X_{i1})$, by contrast, peaks at $X_{i1} = -b/[2(\beta + 1)c]$, below X_{M1} only if the support extends beyond that point. M4 therefore requires $X_{M1} > -b/[2(\beta + 1)c]$ as well as $\beta > 1$. Writing V and C for variances and covariances,

$$C(\Delta x_i, \Delta X_i) = (\beta - 1)^2[bV(X_{i1}) + (\beta + 1)cC(X_{i1}^2, X_{i1})], \quad (17)$$

$$C(x_{i2}, \Delta X_i) = (\beta - 1)\beta[bV(X_{i1}) + c\beta C(X_{i1}^2, X_{i1})], \quad (18)$$

$$V(\Delta X_i) = (\beta - 1)^2V(X_{i1}), \quad (19)$$

$$V(\Delta x_i) = (\beta - 1)^2b^2V(X_{i1}) + (\beta^2 - 1)^2c^2V(X_{i1}^2) + 2bc(\beta - 1)(\beta^2 - 1)C(X_{i1}^2, X_{i1}) + V(u_{i1}) + V(u_{i2}), \quad (20)$$

$$V(x_{i2}) = b^2\beta^2V(X_{i1}) + c^2\beta^4V(X_{i1}^2) + 2bc\beta^3C(X_{i1}^2, X_{i1}) + V(u_{i2}). \quad (21)$$

Since $X_{i1} \geq 0$, $C(X_{i1}^2, X_{i1}) > 0$, and with $c < 0$ the second term in each bracket is negative. Let $\gamma = C(X_{i1}^2, X_{i1})/V(X_{i1})$, the slope of the population regression of X_{i1}^2 on X_{i1} . Then:

- R1. $\rho_{\Delta x \Delta X} > 0$ if and only if $\gamma < -b/[(\beta + 1)c]$.
- R2. $\rho_{x_2 \Delta X} > 0$ if and only if $\gamma < -b/(\beta c)$.
- R3. Since $-b/[(\beta + 1)c] < -b/(\beta c)$, $\rho_{\Delta x \Delta X} > 0$ implies $\rho_{x_2 \Delta X} > 0$, but not the reverse.
- R4. Under M2, R2's condition always holds: $\rho_{x_2 \Delta X} > 0$. The bracket in (18) is the covariance of X_{i1} with $bX_{i1} + c\beta X_{i1}^2$, a function that rises over the whole range of X_{i1} when $\beta X_{M1} \leq -b/2c$; the covariance of a variable with an increasing function of itself is positive. The bracket in (17) is the covariance of X_{i1} with $bX_{i1} + (\beta + 1)cX_{i1}^2$, which rises and then falls, and can be negative.

4.1 A bounded illustration

The piecewise model makes the ceiling's role especially easy to see. For this illustration, X is a nonnegative knowledge index rather than a literal proportion of the domain. Let $X_{i2} = \beta X_{i1}$, with $\beta > 1$, and let $x_{it} = \min(bX_{it}, 1)$. People below $1/(b\beta)$ initially are below the ceiling at both waves, so their observed gain is $b(\beta - 1)X_{i1}$. Those between $1/(b\beta)$ and $1/b$ reach the ceiling only at follow-up: their observed gain is $1 - bX_{i1}$, decreasing with initial knowledge. Those initially at or above $1/b$ begin and end at the ceiling, showing no gain. True gain, $(\beta - 1)X_{i1}$, continues to increase through all three regions.

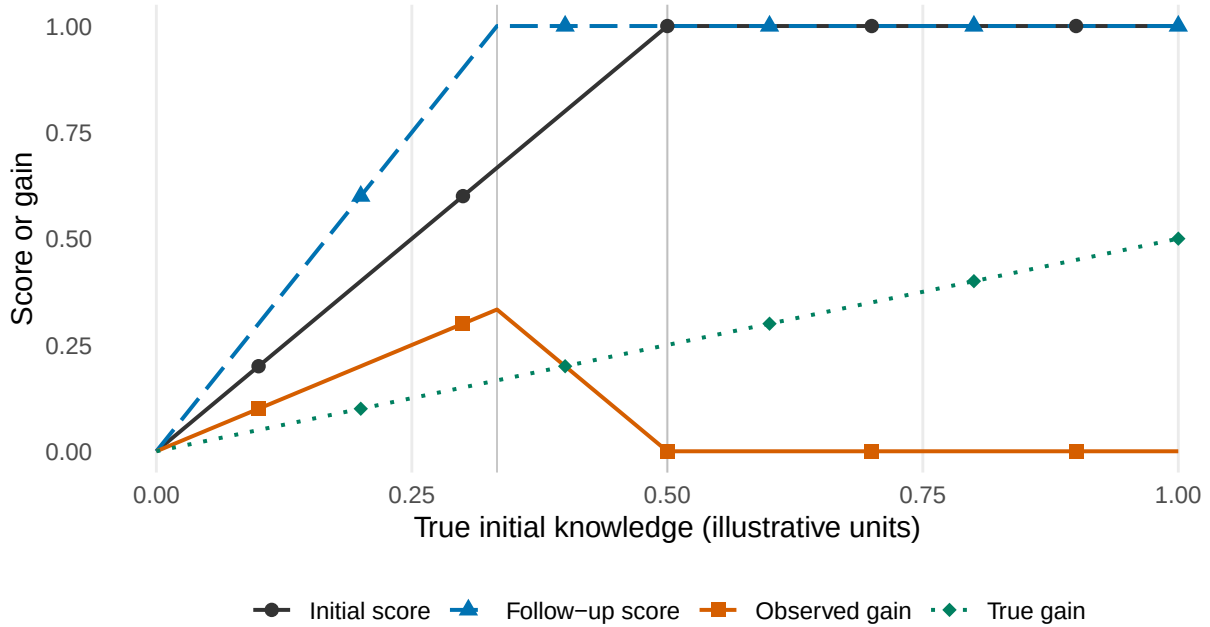


Figure 1: A bounded measurement function with $b = 2$ and proportional learning with $\beta = 1.5$. True initial knowledge runs from zero to one in illustrative units. Vertical lines separate respondents below both ceilings, those reaching the follow-up ceiling, and those at both ceilings. Initial and follow-up scores coincide at one for true initial knowledge of 0.5 or greater.

Observed gain rises, falls, and then becomes constant as true gain increases. Post-process knowledge rises to its ceiling and stays there. It loses the ability to distinguish further gains among high scorers, but it never reverses their ordering. Appendix D gives the moments for a uniform initial distribution.

5 How General Are These Results?

A monotone item-response model extends the argument beyond the linear and quadratic approximations. Let each questionnaire item j have a difficulty d_j on the scale of true knowledge, and let a person know the item with probability $F(X_{it}, d_j)$, nondecreasing in X_{it} . In the simplest case F is a threshold—the person knows the item if and only if $X_{it} > d_j$ —and observed knowledge is $x_{it} = G(X_{it})$, where G is the distribution function of the questionnaire’s difficulties. An easy questionnaire concentrates its difficulties low, so G rises quickly and flattens at 1. A finite bank of threshold items gives a step function G . Uniformly distributed difficulties in a continuous item population give a linear segment bounded by zero and one; a smooth, concentrated distribution can be approximated locally by a quadratic; a logistic F gives an item response model. Let people who do not know an item answer it correctly by lucky guess. Item-response and guessing randomness are independent of initial knowledge and the learning shock, and guessing probabilities do not vary with either. Let true learning be $\Delta X_i = h(X_{i1}, R_i)$, where R_i is a scalar learning shock independent of X_{i1} , and h is nondecreasing in R_i .

Proposition 1. *If h is nondecreasing in X_{i1} (the rich get richer), $\text{Cov}(x_{i2}, \Delta X_i) \geq 0$.*

Proposition 2. *For any h , $\text{Cov}(x_{i2}, \Delta X_i \mid X_{i1}) \geq 0$.*

Proposition 1 strengthens L2: learning must be nondecreasing in initial knowledge at each value of the independent shock, not merely on average. Both propositions follow from the association of independent random variables: any two nondecreasing functions of independent variables have nonnegative covariance (Harris 1960; Esary et al. 1967). With finite second moments, neither argument requires a particular shape of G or distribution of true knowledge. Item-specific guessing probabilities are allowed under the stated independence assumptions (Appendix A). Observed gain has no such unconditional guarantee. Under deterministic measurement, $\Delta x_i = G(X_{i1} + \Delta X_i) - G(X_{i1})$ can rise and then fall in X_{i1} while true gain rises throughout. A ceiling alone does not require this shape over every possible support. With deterministic measurement, conditioning on X_{i1} fixes x_{i1} , so Δx_i and x_{i2} differ only by a constant. Conditioning on the realized observed baseline also makes gain and post-process scores equivalent, but does not condition on true initial knowledge. The two distinctions matter when baseline knowledge is measured with error (Lord 1967; Allison 1990).

The general model establishes nonnegative covariance; the stronger comparison of correlations depends on score variances, as Conditions 3a–3d show. Appendix C derives predictors using both waves and compares fitted item-response estimates with the raw scores.

6 How Large Are the Differences?

The propositions say nothing about magnitudes. To gauge them, we simulate processes and ask how well each proxy tracks true gain. Each simulated process has 300 participants, like a Deliberative Poll. The simulated true-knowledge variable is a right-skewed gamma trait, rather than a literal proportion of all knowable items, and item difficulties are drawn from a range of its quantiles, so that questionnaires run from very easy to moderately hard. People know items either by threshold or with logistic probability, and guess correctly with probability 0, .1 or .25. We vary the number of items (5 to 15), the median learning-rate multiplier and its log-scale dispersion, and three forms of learning: proportional to initial knowledge (L2), unrelated to it, and larger for those who know less.

Within the specified parameter ranges, we retain processes whose observable summaries resemble the polls. For each of 21 Deliberative Polls reported in Luskin et al. (2022), we compute five observable features: the mean and standard deviation of initial observed knowledge, the mean observed gain, the correlation of initial and post-process knowledge, and the correlation of observed gain with initial knowledge. All five empirical features use participants with knowledge scores at both waves. A simulated process passes the filter if each feature falls within the corresponding range across polls; it need not resemble any single poll jointly. Of 4,000 processes with proportional learning, 30% are plausible; with learning unrelated to initial knowledge, 6%; with catch-up learning, 8 of 4,000 processes. Proportional learning passes more often in this grid, but these rates are conditional on the chosen model family and parameter distribution. They do not establish which learning process generated the polls. The gain–initial-knowledge correlation is negative in every poll (-.66 to -.35). Ceilings can produce this pattern, but so can baseline measurement error, because the baseline score is subtracted in observed gain.

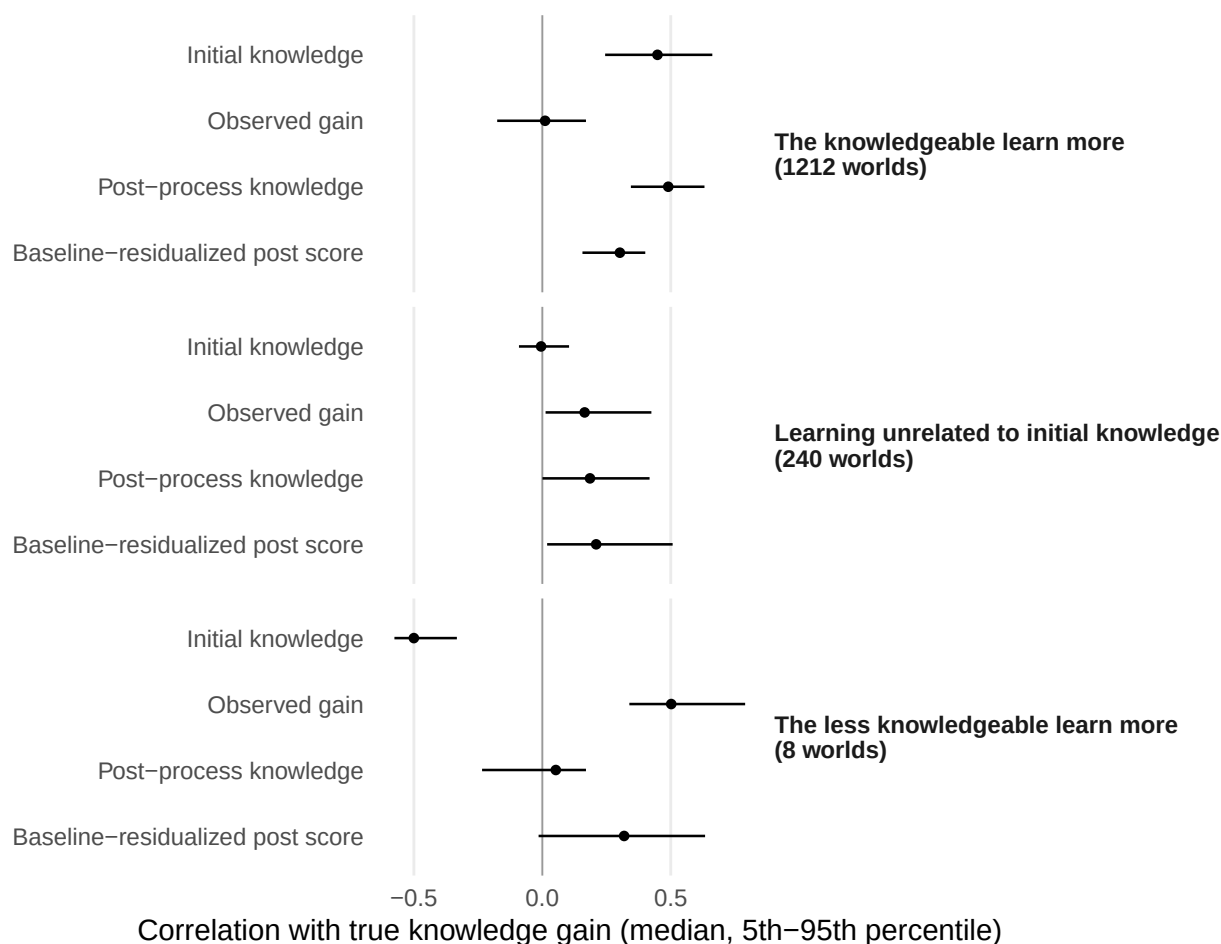


Figure 2: Correlation of each proxy with true knowledge gain across simulated processes whose observable features fall within the marginal ranges across Deliberative Polls. Points are medians and lines span the 5th to 95th percentiles across processes; they describe variation across plausible worlds, not sampling uncertainty. The baseline-residualized score is the residual from regressing post-process on initial knowledge. Every score is correlated with the same, unresidualized true gain.

Figure 2 shows the results. When the knowledgeable learn more, observed gain’s median correlation

with true gain is .01, negative in 46% of plausible processes; post-process knowledge’s is .49, negative in 0% of them, and higher than observed gain’s in 100% of them. Initial knowledge alone has median correlation .45 with true gain. The baseline-residualized post score has median correlation .30 with true gain. The corresponding partial correlation is .34, but answers a different question: it also removes the variation in true gain associated with observed baseline knowledge. When learning is unrelated to initial knowledge, observed gain, follow-up knowledge, and the residualized score have median correlations .16, .19 and .21, respectively; observed gain performs best in the 8 retained catch-up processes. This small selected subset does not characterize catch-up learning generally.

Asking more items helps observed gain only slightly within this simulation grid. Among retained processes with proportional learning, its median correlation is .00 with 5 items and .03 with 15, while post-process knowledge’s is .45 and .55. These are comparisons across separately selected processes, not an experiment holding the process and item difficulty distribution fixed. They do not establish that a longer or better targeted test cannot measure learning well.

6.1 More questions and broader coverage

More questions can reduce response noise; harder questions can reveal learning hidden by ceilings. Item-response scoring and adaptive questioning offer further ways to use responses or target informative items. Agreement between knowledge scores at each wave need not carry over to their differences.

When the knowledgeable learn more, a score recorded before the process can predict subsequent gain. We therefore compare post-process knowledge with initial knowledge as well as observed gain.

Our controlled comparison holds each simulated population and its realized learning fixed while varying questionnaire length from 5 to 30 items and expanding the difficulty range. The comparisons use 100 replications of 1,000 respondents per learning form, without the poll-resemblance filter. Appendix D specifies the populations, response model, and nested questionnaires.

With proportional learning and 10 easy questions, the median correlations with true gain are 0.54 for initial knowledge, -0.14 for observed gain, and 0.50 for follow-up knowledge. With 30 questions covering the broader range, observed gain’s correlation is 0.00, and follow-up knowledge’s is 0.64. In the easy proportional-learning scenario, initial knowledge predicts true gain at least as well as the follow-up score. The ordering changes with the learning process: under catch-up learning, the 30-item broad questionnaire gives correlations of 0.80 for observed gain and 0.39 for follow-up knowledge. More easy questions can make a negatively correlated gain score more precise without making it a better proxy.

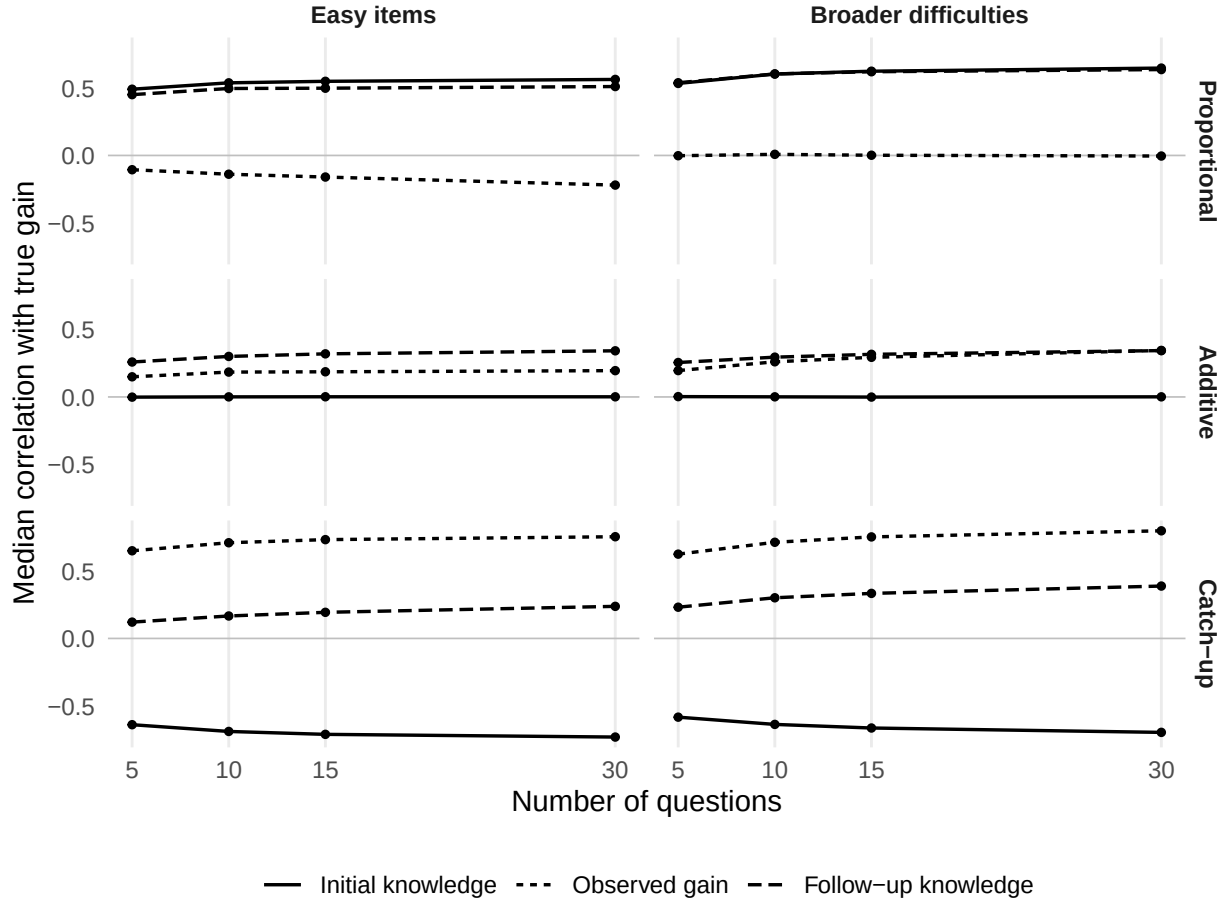


Figure 3: Questionnaire length and difficulty coverage in paired simulations. Lines show median correlations over 100 replications of 1,000 respondents for each learning form. Every questionnaire design within a replication measures the same people and realized gains. Shorter questionnaires are nested in the 30-item bank.

6.2 Different people can learn different things

Learning in a Deliberative Poll may differ across people partly because the information provided differs across small groups. Some learn one set of facts, others another. Selective learning can have the same consequence even under common exposure: people attend to different topics or different parts of a discussion. A questionnaire drawn from one part of the discussion can miss learning elsewhere.

A two-domain simulation shows how uneven coverage affects the scores. Initial domain knowledge follows independent gamma distributions, the domains receive equal weight, and total learning is proportional to mean initial knowledge. Each person receives all new information in one randomly assigned domain; the amount added preserves the same total gain regardless of its allocation. Both questionnaires have 10 items: one draws five from each domain, the other all ten from the first. Item difficulties cover the broader range. Across 100 replications, balanced and one-domain questionnaires have median follow-up correlations with total gain of 0.53 and 0.42, respectively. Among people exposed only to the second domain, the one-domain follow-up score still correlates

0.39 with total gain. The post-process score predicts total learning here through prior knowledge, even though its items cover none of the material learned. Appendix D reports the domain-specific results.

7 Does the Choice Matter?

Consider a question the literature on deliberation asks often: who learns? In the same 21 polls, we regress each learning proxy on whether education is strictly above the poll’s participant median, poll by poll, and pool the estimates in a Bayesian random-effects meta-analysis (Röver 2020). With observed gain, the above-median education difference is uncertain (.017 [-.005, .041]), with a negative estimate in 6 of the polls. With post-process knowledge given initial knowledge, the above-median group scores higher (.066 [.049, .085]), with a negative estimate in 1 of 21 polls (Figure 4). Post-process knowledge alone, without initial knowledge, gives .116 [.098, .134], mixing what people learned with what they already knew. Because observed gain is follow-up minus baseline knowledge, using gain with baseline held constant gives exactly the same education coefficient as using follow-up with baseline held constant. Thus baseline adjustment accounts for the difference between the two panels of Figure 4.

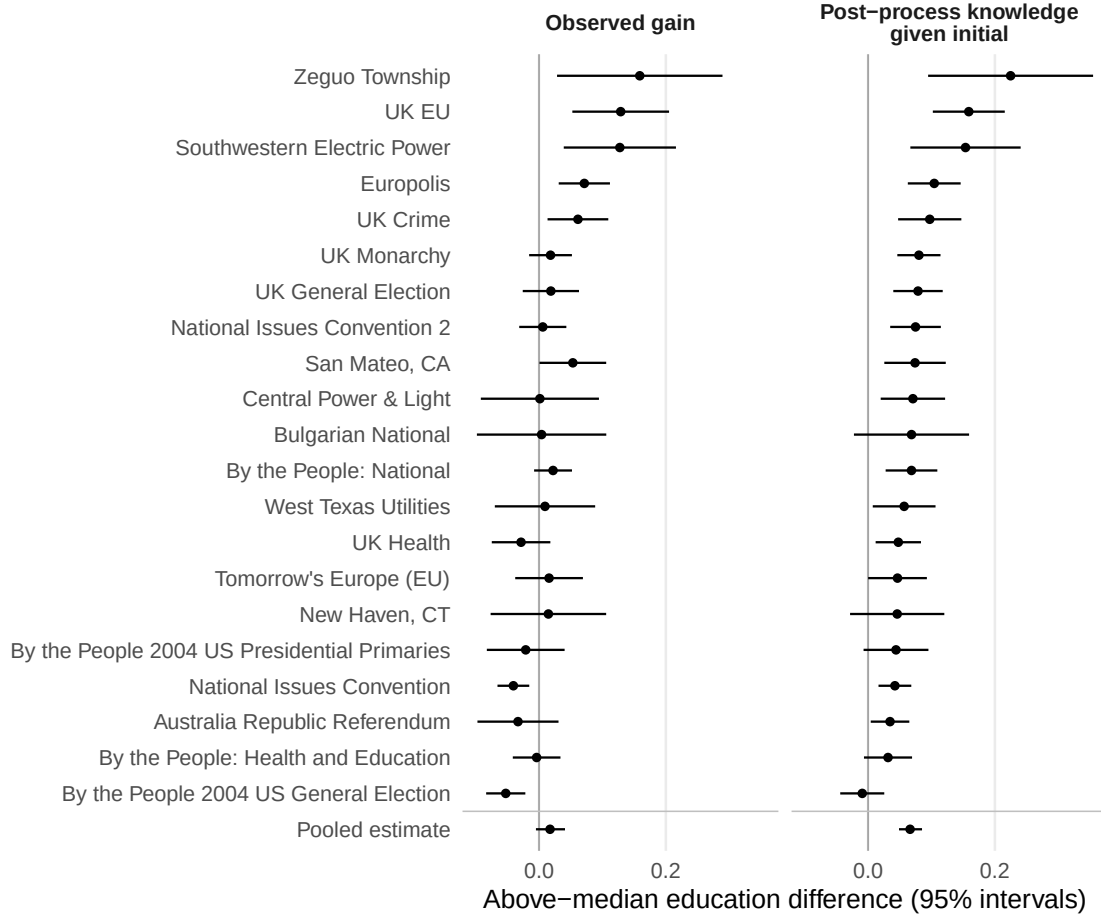


Figure 4: Difference in learning proxies for participants above versus at or below their poll’s education median, by poll, with 95% intervals. Left: observed gain. Right: post-process knowledge holding initial knowledge constant. Standard errors are clustered by small group; the pooled estimates are from Bayesian random-effects meta-analyses.

The education indicator comes from dp-data’s reviewed poll-specific ordered measures. Its fixed reference population is the unique historical-polardata participant sample with observed education, before exclusions for outcomes or group IDs; ties at the median remain in the lower group. We use the exported indicator without recomputing cutoffs. Education, paired knowledge and discussion-group membership must be observed for a participant to enter the regressions. The comparisons are associations with observed scores, not causal effects of education or direct measurements of true learning. Poll estimates use HC1 standard errors clustered by discussion group and normal 95% intervals. The meta-analysis uses a half-normal prior with scale .1 on between-poll heterogeneity and an improper uniform prior on the pooled mean; its intervals are 95% posterior credible intervals.

Learning is often also used to explain attitude change: in Deliberative Polls, the participants who learn the most have been expected to account for much of the net change in opinion. We estimate these associations on all 129 attitude indices in the corrected data, regressing the change in each index on a learning proxy and the participant’s distance from the other complete-case members of the small group at baseline. Records with missing scores or group IDs and groups with fewer than two complete cases are excluded. The net direction and proxy standard deviation are computed

within that same analysis sample. The share of indices on which the coefficient is positive is 50% for observed gain, 59% for post-process knowledge and 53% for post-process knowledge given initial knowledge; the shares significantly positive are 5%, 16% and 12%, respectively (Appendix Table B2). These summaries show proxy-specific associations; they do not establish that learning has no relationship with attitude change. Initial knowledge also has positive associations. The tests are unadjusted for multiple comparisons, indices within a poll are dependent, and the direction is estimated from the same sample. The significance shares therefore are descriptive rather than a calibrated test against chance.

7.1 Predicting gain on other questionnaire items

Observed gain on disjoint questionnaire items provides an empirical target for comparing the scores. We divide each poll’s matched items into predictor and validation sets, with half, rounded down, assigned to prediction and at least two questions in each set. Separating the sets prevents shared responses from creating agreement mechanically. This comparison targets gain on other tested items; topic-wide gain remains unobserved.

The comparison uses 5,813 participants with complete paired item responses across 21 polls. We repeat disjoint item partitions and five-fold respondent splits 100 times, keeping discussion groups together. The easier-item partitions choose predictor items from training respondents’ baseline answers; residualization is also fitted only on training respondents. All scores within a test fold use the same respondents and validation items. Figure 5 reports paired correlation differences. Appendix D gives the sample and estimation details.

When questions are randomly divided into two sets, observed gain on one set correlates more strongly with observed gain on the other than the follow-up score does in 16 of 21 polls. The median across polls of the post-process-minus-gain correlation difference is -0.04. Selecting easier predictor items changes the difference to 0.01, with observed gain ahead in 9 polls. This selection also leaves harder validation items, changing both sides of the comparison. Post-process knowledge improves on initial knowledge alone under both partition rules, by median correlation differences of 0.18 and 0.17, respectively. These correlations compare scores with observed gain on other questions, not with independently measured true learning.

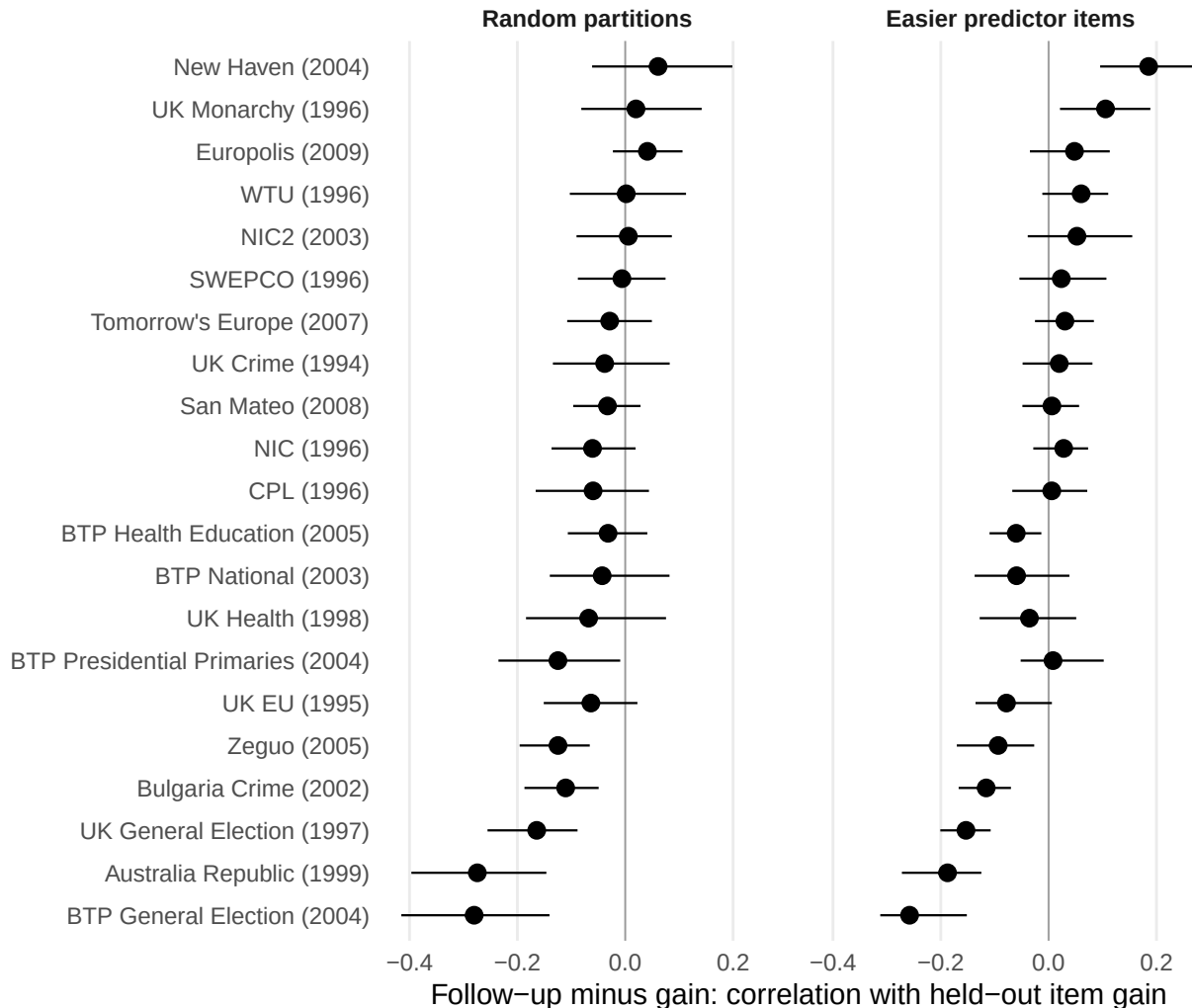


Figure 5: Correlations of follow-up scores and observed gains on one question set with observed gains on a separate set. Each point is the median, across 100 splits, of the within-split median paired correlation difference across five test folds. Lines span the 5th–95th percentiles across splits, not sampling confidence intervals. Random partitions are compared with selecting easier predictor items using only training respondents’ baseline answers. Positive differences favor follow-up scores. All methods within a fold use the same complete paired-item sample.

8 Conclusion

When the knowledgeable learn more, easy questionnaires can make the greatest learners appear to gain the least. We derive conditions under which post-process knowledge correlates more strongly with true gain than observed gain does. The advantage holds in 100% of the retained proportional-learning simulations, with median correlations of .49 and .01. The poll analyses show why the choice matters: scoring and baseline adjustment change associations with education and attitude change.

But observed gain performs better in the catch-up simulations. With randomly divided question sets, observed gain on one set usually correlates more strongly with observed gain on the other than

the follow-up score does. Initial knowledge itself predicts gain when the knowledgeable learn more, even when a questionnaire misses the material learned. Estimating an amount requires a defined knowledge domain and a common scale across waves. The joint item-response comparison shows that using both waves can improve recovery, with performance depending on the learning model and the information supplied by the items.

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Appendix

A Proofs

A.1 Result 1 and Condition 1a

The correlation has the sign of its covariance. By (9),

$$\begin{aligned}\rho_{\Delta x \Delta X} < 0 &\iff (\beta - 1)(b_2\beta - b_1)\sigma_{X_1}^2 + b_2\sigma_v^2 < 0 \\ &\iff (\beta - 1)(b_1 - b_2\beta)\sigma_{X_1}^2 > b_2\sigma_v^2.\end{aligned}$$

The last inequality is Condition 1a. Its left side is positive by (7) and (8), so the covariance can be negative when the learning disturbance has sufficiently small variance.

A.2 Result 2

By (10),

$$\sigma_{x_2 \Delta X} = (\beta - 1)b_2\beta\sigma_{X_1}^2 + b_2\sigma_v^2 > 0.$$

Both addends are positive by (6) and (7). The correlation is therefore positive too.

A.3 Result 3 and Conditions 3a–3d

First, post-process knowledge always has the larger covariance with true gain:

$$\sigma_{x_2 \Delta X} - \sigma_{\Delta x \Delta X} = (\beta - 1)b_1\sigma_{X_1}^2 > 0.$$

If the gain covariance is nonpositive, Result 2 gives the desired correlation ordering immediately. If it is positive, the ordering also follows whenever post-process knowledge has the smaller variance. Thus

$$\sigma_{x_2}^2 < \sigma_{\Delta x}^2 \implies \rho_{x_2 \Delta X} > \rho_{\Delta x \Delta X}.$$

Conditions 3a and 3b. Subtracting (12) from (11) gives

$$\begin{aligned}\sigma_{\Delta x}^2 - \sigma_{x_2}^2 &= [(b_2\beta - b_1)^2 - b_2^2\beta^2]\sigma_{X_1}^2 + \sigma_{u_1}^2 \\ &= (b_1 - 2b_2\beta)b_1\sigma_{X_1}^2 + \sigma_{u_1}^2.\end{aligned}$$

Positivity is Condition 3a. Condition 3b makes the first term nonnegative; the strictly positive measurement-error variance makes the sum positive.

Condition 3c. Condition 1a and Result 2 imply

$$\rho_{\Delta x \Delta X} < 0 < \rho_{x_2 \Delta X}.$$

Condition 3d. Since $\Delta x_i = x_{i2} - x_{i1}$,

$$\begin{aligned}\sigma_{\Delta x}^2 - \sigma_{x_2}^2 > 0 &\iff \sigma_{x_1}^2 - 2\sigma_{x_1 x_2} > 0 \\ &\iff \gamma_{21} = \frac{\sigma_{x_1 x_2}}{\sigma_{x_1}^2} < \frac{1}{2}.\end{aligned}$$

A.4 Guessing

Write $k_i = \kappa r_i$. The score and gain become

$$y_{i2} = (1 - k_i)x_{i2} + k_i, \quad \Delta y_i = (1 - k_i)\Delta x_i.$$

Independence of k_i from the knowledge scores and true gain gives

$$\begin{aligned} \text{Cov}(\Delta y_i, \Delta X_i) &= E(1 - k_i) \text{Cov}(\Delta x_i, \Delta X_i), \\ \text{Cov}(y_{i2}, \Delta X_i) &= E(1 - k_i) \text{Cov}(x_{i2}, \Delta X_i). \end{aligned}$$

The multiplier is nonnegative and is positive unless everyone guesses correctly with probability one. The strict signs of Results 1 and 2 survive whenever it is positive.

A.5 Propositions 1 and 2

Fix the questionnaire's item difficulties. Let $m(s)$ be the expected time-2 score at true knowledge s , averaging over item-response and guessing randomness that is independent of (X_{i1}, R_i) . Under the monotone item-response model, m is nondecreasing. Define

$$q(x, r) = m(x + h(x, r)).$$

Because true gain is determined by (X_{i1}, R_i) , the law of total covariance gives

$$\text{Cov}(x_{i2}, \Delta X_i) = \text{Cov}(q(X_{i1}, R_i), h(X_{i1}, R_i)).$$

Proposition 1. If h is nondecreasing in both arguments, so is q . The variables X_{i1} and R_i are independent and hence associated (Esary et al. 1967). Two nondecreasing functions of associated variables have nonnegative covariance. Therefore

$$\text{Cov}(x_{i2}, \Delta X_i) \geq 0.$$

Proposition 2. Conditional on $X_{i1} = x$, both $q(x, R_i)$ and $h(x, R_i)$ are nondecreasing functions of the scalar R_i . Their covariance is nonnegative (Harris 1960), giving

$$\text{Cov}(x_{i2}, \Delta X_i \mid X_{i1} = x) \geq 0.$$

This proof does not require h to be nondecreasing in its first argument.

The conditioning variable in Proposition 2 is true initial knowledge. The result does not guarantee a nonnegative partial correlation after adjusting for noisy observed initial knowledge: that partial correlation is negative in 1 of the 1,212 retained proportional-learning simulations.

B The polls and further results

Table B1: Observable features of the Deliberative Polls. The first five judge which simulated processes are plausible; γ_{21} is the slope of the regression of x_2 on x_1 (Condition 3d).

Poll	Mean x_1	Mean gain	SD x_1	$r(x_1, x_2)$	$r(\Delta x, x_1)$	γ_{21}
San Mateo, CA	.15	.17	.14	.11	-.46	.18
Europolis	.20	.17	.16	.21	-.46	.29
Central Power & Light	.29	.16	.19	.27	-.62	.26
Zeguo Township	.31	.11	.31	.30	-.50	.35
By the People: Health and Education	.35	.06	.21	.35	-.55	.36
Tomorrow's Europe (EU)	.37	.18	.19	.47	-.44	.52
New Haven, CT	.37	.23	.18	.39	-.35	.53
National Issues Convention 2	.37	.12	.17	.56	-.44	.58
By the People: National	.39	.03	.18	.61	-.45	.60
West Texas Utilities	.40	.29	.24	.39	-.56	.38
Bulgarian National	.43	.09	.25	.29	-.58	.30
Southwestern Electric Power	.44	.20	.23	.22	-.57	.25
National Issues Convention	.48	.10	.20	.62	-.59	.52
Australia Republic Referendum	.48	.28	.23	.59	-.66	.46
By the People: US Primaries, 2004	.51	.12	.28	.72	-.42	.69
UK EU	.52	.20	.29	.45	-.52	.45
UK Crime	.52	.09	.23	.61	-.36	.67
UK General Election	.54	.10	.21	.53	-.52	.51
UK Monarchy	.65	.15	.19	.56	-.42	.60
UK Health	.66	.08	.23	.56	-.42	.59
By the People: US Election, 2004	.75	.07	.24	.71	-.51	.63

Table B2: Learning and attitude change. Each attitude index's change is regressed on the proxy and the distance from the small group's mean; the proxy's coefficient is signed so that positive means moving with the sample's net change. Significance at the 5% level with standard errors clustered by small group.

Proxy	Indices	Positive	Sig. positive	Sig. negative
Observed gain	129	.50	.05	.05
Post-process knowledge	129	.59	.16	.06
Post-process knowledge given initial	129	.53	.12	.04
Initial knowledge	129	.59	.10	.09

C Alternatives to post-process knowledge

C.1 Target and comparison criteria

The positive covariance of post-process knowledge with true gain does not establish that it is the best predictor. Let $\Delta = X_2 - X_1$ be true gain on a fixed knowledge scale and let $T_t = x_t$ denote the observed score at wave t , suppressing person subscripts. We compare predictors by their Pearson correlation with the same Δ and, when predictions are expressed in its units, by mean squared error $R(f) = E\{(\Delta - f)^2\}$. Assume finite second moments and positive variances wherever a correlation or variance ratio is used. Rank correlation is a separate criterion; the Pearson results below do not establish optimality for ranks.

The formal comparisons distinguish population predictors, whose defining moments or conditional distributions are known, from estimates obtained by fitting a model. The former establish how much information could help. The latter require identification and can lose that advantage through estimation error or misspecification.

C.2 Combining both scores in the linear model

Let $L_2 = a + bT_2$ be the best affine predictor of Δ using the follow-up score and let $L_{12} = c + w_1T_1 + w_2T_2$ be the best affine predictor using both scores. Allowing the baseline coefficient to be zero nests L_2 in the second class, so $R(L_{12}) \leq R(L_2)$. Calibrating both predictors to Δ makes this an error comparison in common units.

To characterize the gain, define the baseline residual after linear prediction from follow-up knowledge:

$$Z = (T_1 - ET_1) - \frac{\text{Cov}(T_1, T_2)}{\text{Var}(T_2)}(T_2 - ET_2).$$

Then $EZ = 0$ and $\text{Cov}(Z, T_2) = 0$. Orthogonality gives

$$L_{12} = L_2 + \frac{\text{Cov}(\Delta, Z)}{\text{Var}(Z)}Z, \tag{22}$$

$$R(L_2) - R(L_{12}) = \frac{\text{Cov}(\Delta, Z)^2}{\text{Var}(Z)}. \tag{23}$$

For a direct proof, put $q = \text{Cov}(\Delta, Z)$ and expand the risk of $L_2 + kZ$. Since $\text{Cov}(L_2, Z) = 0$,

$$R(L_2 + kZ) = R(L_2) - 2kq + k^2 \text{Var}(Z).$$

Its minimizer is $k = q/\text{Var}(Z)$, proving both expressions. Baseline strictly improves linear prediction if and only if $q \neq 0$.

The corresponding correlation comparison is

$$\text{Corr}(L_{12}, \Delta)^2 - \text{Corr}(T_2, \Delta)^2 = \frac{\text{Cov}(\Delta, Z)^2}{\text{Var}(Z) \text{Var}(\Delta)} \geq 0. \tag{24}$$

To obtain it, use $\text{Cov}(L_{12}, \Delta) = \text{Var}(L_{12})$ and $\text{Var}(L_{12}) = \text{Var}(L_2) + q^2/\text{Var}(Z)$. Under the paper's linear-model assumptions, T_2 has positive covariance with Δ , so the comparison also orders the unsquared correlations.

Using the notation of the linear model, let $z = (T_1 - ET_1, T_2 - ET_2)^\top$, $\Sigma = \text{Var}(z)$, and $c_\Delta = \text{Cov}(z, \Delta)$. The same predictor can be written

$$L_{12} = E\Delta + c_\Delta^\top \Sigma^{-1} z, \quad (25)$$

where

$$c_\Delta = \begin{pmatrix} b_1(\beta - 1)\sigma_{X_1}^2 \\ b_2\{\beta(\beta - 1)\sigma_{X_1}^2 + \sigma_v^2\} \end{pmatrix}.$$

The positive measurement-error variances and zero covariance restrictions make Σ positive definite. Writing $D = \det(\Sigma) > 0$, substitution yields

$$w_1 = \frac{b_1\sigma_{X_1}^2\{(\beta - 1)\sigma_{u_2}^2 - b_2^2\sigma_v^2\}}{D}. \quad (26)$$

Thus the baseline weight is positive when $(\beta - 1)\sigma_{u_2}^2 > b_2^2\sigma_v^2$ and negative when the inequality reverses. When baseline knowledge predicts learning and follow-up measurement error is relatively large, baseline helps recover knowledge associated with greater learning. When variation in the learning disturbance is relatively large, subtracting baseline helps distinguish newly acquired knowledge from knowledge already possessed. Neither the fixed weight of -1 in observed gain nor the zero weight in the follow-up score is generally optimal. Residualizing follow-up knowledge on baseline imposes another particular pair of weights and has no general advantage over T_2 for predicting the same true gain.

Equation (25) is a population benchmark. The two observed scores alone do not identify its measurement slopes, error variances, and learning moments. Repeated indicators, external calibration, or explicit sensitivity assumptions must supply that information. No normality assumption is needed for the best affine predictor; interpreting it as the full conditional mean requires more structure.

C.3 Conditional means using item responses

Write $\mathcal{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$ for all item responses at both waves. The population predictor with minimum mean squared error is

$$m_{12} = E[\Delta \mid \mathcal{Y}]. \quad (27)$$

For any square-integrable predictor $f(\mathcal{Y})$,

$$R(f) = R(m_{12}) + E\{[f(\mathcal{Y}) - m_{12}]^2\}. \quad (28)$$

Indeed, expanding the square leaves a cross term whose expectation is zero because $E(\Delta - m_{12} \mid \mathcal{Y}) = 0$. The minimizer is therefore m_{12} , uniquely up to events of probability zero.

Let $m_2 = E[\Delta \mid \mathbf{Y}_2]$ use only follow-up item responses and let $m_T = E[\Delta \mid T_2]$ use only their aggregate score. These predictors obey

$$R(m_{12}) \leq R(m_2) \leq R(m_T) \leq R(L_2).$$

Each step enlarges the information or function class available for prediction. In particular,

$$R(m_2) - R(m_{12}) = E[(m_{12} - m_2)^2], \quad (29)$$

$$R(m_T) - R(m_2) = E[(m_2 - m_T)^2]. \quad (30)$$

Baseline adds value exactly when it changes expected gain after conditioning on all follow-up responses. Item patterns add value beyond the follow-up total exactly when they change expected gain at a given total. If the total retains all the relevant information in the follow-up pattern, the second improvement is zero.

The conditional mean also maximizes squared Pearson correlation with true gain. For any square-integrable, nonconstant score $s(\mathcal{Y})$,

$$\text{Cov}(s, \Delta) = \text{Cov}(s, m_{12}).$$

Cauchy–Schwarz and $\text{Cov}(m_{12}, \Delta) = \text{Var}(m_{12})$ give

$$\text{Corr}(s, \Delta)^2 \leq \frac{\text{Var}(m_{12})}{\text{Var}(\Delta)} = \text{Corr}(m_{12}, \Delta)^2. \quad (31)$$

The final equality presumes m_{12} is nonconstant. If it is constant, every such score has zero covariance with gain. Positive affine transformations of a nonconstant m_{12} attain the same positive correlation, while only m_{12} itself minimizes squared error on the specified gain scale.

Estimating the two knowledge levels from the same joint posterior and then subtracting is equivalent to integrating gain:

$$E[X_2 \mid \mathcal{Y}] - E[X_1 \mid \mathcal{Y}] = E[X_2 - X_1 \mid \mathcal{Y}].$$

Scoring each level from its own wave instead gives $E[X_2 \mid \mathbf{Y}_2] - E[X_1 \mid \mathbf{Y}_1]$, which generally differs from m_{12} and has no general advantage over the follow-up score. Transforming posterior mean abilities before differencing also generally differs from integrating gain when the transformation to knowledge is nonlinear.

These results concern the true joint distribution. To see why a fitted model need not inherit them, let $\widehat{m}_{12}$ and $\widehat{m}_2$ be trained on data independent of the scored respondent. Averaging over training and test data, (28) implies

$$\begin{aligned} R(\widehat{m}_{12}) - R(\widehat{m}_2) &= -E[(m_{12} - m_2)^2] \\ &\quad + E[(\widehat{m}_{12} - m_{12})^2] - E[(\widehat{m}_2 - m_2)^2]. \end{aligned}$$

The information benefit can be outweighed by additional estimation error or misspecification. The conditional-mean theorem therefore supplies a benchmark, not a guarantee for a particular fitted IRT model.

C.4 What the assumptions identify

Under the general model, the response function is only required to be monotone. Rescaling true knowledge and the item difficulties by the same positive constant leaves every response probability unchanged while rescaling true gain. Thus the observations do not identify the amount learned without a scale calibration. Under proportional learning, even an increasing power transformation preserves the rich-get-richer structure:

$$X_{i2} = X_{i1}(1 + R_i) \implies X_{i2}^p - X_{i1}^p = X_{i1}^p[(1 + R_i)^p - 1], \quad p > 0.$$

Transforming the response function accordingly leaves the response distribution unchanged. Different powers can change which of two people with different initial knowledge and learning shocks gained more. For example, changes from 1 to 2 and from 3 to 3.5 rank as $1 > .5$ on the original scale, but as $3 < 3.25$ on the squared scale. A monotone knowledge scale therefore need not identify rankings of individual gains either.

This limitation is especially consequential for the proportion of *all* relevant facts known. A short, deliberately easy questionnaire does not identify its relation to that universe. The gamma trait used in the simulations supplies a scale by assumption. Its gains are meaningful within those simulated worlds, but are not estimates of the fraction of all political knowledge acquired by poll participants.

C.5 Guessing-adjusted follow-up knowledge

Suppose the same probability $g < 1$ of a correct guess applies to every person and item. The usual correction to the observed follow-up proportion y_{i2} is

$$\tilde{x}_{i2} = \frac{y_{i2} - g}{1 - g}.$$

It changes the units and level, but is a positive affine transformation. Therefore it has exactly the same ranking and correlation with true gain as y_{i2} . In our simulations, each process has one common guessing probability, so this correction cannot improve the follow-up score's correlation. Nor does it remove the ceiling. The statement concerns ranking and correlation, not bias in the mean proportion of questionnaire items known.

Item-specific or person-specific guessing probabilities can change the ranking. If Y_{ij2} is item correctness and $g_{ij} < 1$ the probability of guessing correctly when the item is not known, the corrected contribution is

$$\tilde{K}_{ij2} = \frac{Y_{ij2} - g_{ij}}{1 - g_{ij}}.$$

With a correct answer certain when the item is known, this contribution has conditional expectation equal to the item's binary knowledge state. When the item is unknown its conditional variance is $g_{ij}/(1 - g_{ij})$, so removing guessing bias can amplify noise. Estimated guessing probabilities add uncertainty. Negative corrected contributions should not simply be truncated to zero, which would undo the unbiasedness argument. If guessing varies with initial knowledge or the learning shock, the propositions' independence assumptions also need reconsideration. The item transition model of Cor and Sood (2016) uses both waves to estimate learning about the questions asked under a no-forgetting restriction; it does not by itself identify each person's topic-wide learning.

C.6 A joint item-response model

A joint IRT model fits the two waves together and estimates change directly. Let Y_{ijt} indicate a correct response and θ_{it} denote knowledge on a common latent scale. For example,

$$\Pr(Y_{ijt} = 1 \mid \theta_{it}) = g_j + (1 - g_j) \text{logit}^{-1}[a_j(\theta_{it} - d_j)], \quad (32)$$

where $a_j > 0$ is discrimination, d_j difficulty, and g_j a lower asymptote. Common anchor items link the scale across waves; their response parameters must be invariant, or changes in item functioning must be modeled. A joint distribution for $(\theta_{i1}, \theta_{i2})$ lets both waves inform each person’s change. Measurement invariance is a substantive requirement for interpreting latent change (Millsap 2010; Qian 2018). Responses to the same item across waves may remain dependent even after conditioning on ability, so local independence also needs checking.

Let $X_{it} = \psi(\theta_{it})$ express knowledge on the target scale. The fitted model supplies

$$\widehat{\Delta X}_i = \int \int [\psi(\theta_2) - \psi(\theta_1)] \hat{p}(\theta_1, \theta_2 \mid y_i) d\theta_1 d\theta_2.$$

This integrates gain itself. For nonlinear ψ , it differs from transforming two posterior mean abilities and subtracting them. A known ψ is an identifying assumption; IRT alone cannot recover the relation to the proportion of all relevant facts known. Uncertainty must account for the posterior dependence between the two abilities and, for full inference, uncertainty in the item and population parameters. Fitting each wave separately, standardizing both to mean zero, and subtracting the scores would erase average change. Imposing nonnegative individual change or proportional learning in the prior would build that restriction into the answer; the data should also be fitted with a model allowing forgetting and different relations between baseline knowledge and gain.

A latent-standard-deviation change is not a percentage of knowledge. For an interpretable amount, define a fixed reference set of items and weights $\omega_j \geq 0$ summing to one. Under the knowledge-versus-guessing interpretation of (32), define

$$K(\theta) = \sum_j \omega_j \text{logit}^{-1}[a_j(\theta - d_j)].$$

Then estimate $E\{K(\theta_{i2}) - K(\theta_{i1}) \mid \mathbf{Y}_{i1}, \mathbf{Y}_{i2}\}$ and its interval. This expresses change in the expected proportion of the reference items known, with guessing excluded. The reference items and weights stay fixed across waves. Unlike a raw ability difference, this quantity is unchanged by rescaling the IRT ability and item parameters together. Generalization to a larger topic requires a defensible item-sampling frame or external calibration; the tested items alone cannot represent everything there is to know. A generic IRT lower asymptote also need not be literal random guessing, so that interpretation requires its own evidence.

The model should use incorrect and explicit “don’t know” answers separately when the response process supports that distinction, while treating absent questionnaires as missing. A freely estimated guessing parameter for every person is unlikely to be well determined by a short battery; item-level guessing parameters, constrained by the answer options or external calibration, are a more defensible starting point. To study who learns, education and other predictors can enter the joint change model, with measurement invariance checked across groups. Regressing separately shrunken person scores as though they were observed learning would discard their uncertainty and can distort those associations.

C.7 A head-to-head comparison

High agreement between IRT and conventional *level* scores cannot establish that their estimates of *change* perform alike. The relevant test is recovery of true gain on independent data, including respondents whose initial scores are already at the ceiling.

We implement (27) with a joint model for baseline log knowledge and its change. The baseline has mean zero and variance one, fixing the scale, and its change has an estimated mean, variance, and covariance with baseline. The pair has a working bivariate normal distribution. Knowledge is $X = \exp(\theta)$, a positive trait on this normalized scale, not the proportion of all knowable facts. Item slopes and intercepts are estimated subject to equality across waves; guessing probabilities are supplied by the test design. Slopes are constrained to $[0.2, 5]$ and change variance to be at least .0025. We use marginal likelihood estimation (Chalmers 2012), expanding the integration range when the fitted change distribution extends beyond it, followed by Gaussian quadrature to integrate gain. These are empirical Bayes estimates: the fitted item and population parameters are treated as known when scoring a new person.

Each replication uses 300 training respondents and 1,000 independent test respondents answering the same items. Baseline log knowledge is standard normal. Proportional, additive, and catch-up learning use the same positive lognormal rate mechanism as the earlier simulations, with median .5 and log-scale standard deviation .5. The main scenarios have 5, 10, or 15 easy items; further scenarios broaden the difficulty range, vary guessing across items, or make the follow-up items easier without telling the fitted model. Unlike the earlier grid, this comparison does not select simulations for resemblance to the polls. All predictions target the same realized $X_{i2} - X_{i1}$ in a given replication. The fitted normal distribution of log change is an approximation even in the proportional-learning scenarios, rather than access to the true simulation law.

We compare seven procedures. Initial knowledge supplies a baseline-only benchmark. Three further procedures are the original scores: raw gain, raw follow-up knowledge, and guessing-adjusted follow-up knowledge. The other three estimate an amount in the specified trait units: the joint posterior mean gain; the posterior mean gain using only follow-up responses; and the difference between posterior mean knowledge levels scored from each wave separately. The latter two use the same fitted model as the joint estimate, so this comparison holds item calibration fixed and changes how the response information is used. The fitted model sees no true knowledge or true gain. Raw questionnaire proportions and the positive knowledge trait have different units, so we compare all seven by correlation and assess squared error only for the three estimates expressed in the target units.

Of 80 attempted fits, 75 yield converged, numerically stable models. Tables C1 and C2 compare methods on those same successful replications, with failures included in each scenario’s fit count. The reported summaries are medians across training/test replications, not confidence intervals. The repository also records rank correlations, mean error, conditional interval coverage, and results for respondents who answered every baseline item correctly.

Table C1: Correlation with true gain in independent test respondents. Entries are median correlations across successful replications; Fits reports successful fits over all attempts. Learning is proportional except in the additive and catch-up rows. All scenarios use 10 items unless stated otherwise.

Scenario	Fits	Initial	Gain	Follow-up	Guess adj.	Joint IRT
Easy, 5 items	10/10	.36	-.06	.35	.35	.44
Easy, 10 items	10/10	.44	-.09	.41	.41	.55
Easy, 15 items	10/10	.46	-.12	.43	.43	.63
Broader difficulties	10/10	.50	-.02	.49	.49	.68
Additive learning	10/10	.01	.12	.23	.23	.07
Catch-up learning	5/10	-.58	.62	.18	.18	.53
Item-specific guessing	10/10	.44	-.10	.41	.39	.55
Easier follow-up items	10/10	.40	-.19	.33	.33	.52

Table C2: Root mean squared error for estimates of the amount learned. Lower is better. All three procedures use the same estimated model and predict gain in the same positive-trait units. Entries are medians over the successful replications in the correlation table. Raw score proportions are not in these units and are not assigned an RMSE.

Scenario	Joint gain	Follow-up only	Separate-wave difference
Easy, 5 items	1.72	1.65	2.18
Easy, 10 items	1.18	1.23	1.88
Easy, 15 items	1.17	1.24	1.96
Broader difficulties	1.09	1.19	2.04
Additive learning	0.76	0.61	1.28
Catch-up learning	1.78	2.05	1.76
Item-specific guessing	1.54	1.52	2.13
Easier follow-up items	2.16	2.27	2.86

With 10 easy items and proportional learning, the joint estimate has median correlation 0.55 with true gain, compared with 0.41 for the follow-up score and -.09 for raw gain. Initial knowledge alone correlates 0.44 with gain, again exceeding the follow-up score. The joint estimate improves on that baseline benchmark as well. Its median RMSE is 1.18, versus 1.23 using only follow-up responses; it has lower RMSE in 9 of the 10 paired replications. More items and a broader difficulty range improve recovery. With only five easy items, however, the joint estimate’s higher correlation does not translate into lower median squared error than the follow-up-only model.

The departures matter. Under additive learning, the fitted joint estimator recovers little of the variation in true gain. Under catch-up learning, raw gain has the larger correlation, and the fitting software rejects some replications because a follow-up item is answered correctly by every training respondent. Making the follow-up items easier produces positive bias of 0.93 trait units and reduces joint conditional interval coverage to 75.8 percent. These intervals condition on estimated parameters; they are not full confidence intervals. Item-specific guessing adjustment also performs slightly worse than the raw follow-up score in this design.

Thus the conditional mean is the optimal target under squared loss, but this fitted normal-change implementation is not uniformly best. The results support using both waves when the measurement

and learning models are adequate. They also make flexible change distributions, item-invariance checks, and harder questions substantive parts of estimation. Ten replications per scenario provide an initial comparison, not a precise estimate of performance differences or a selection of one model for the polls.

C.8 What to do at the ceiling

IRT cannot create information where the questions supply almost none. When a participant answers every easy item correctly, large ranges of high ability have almost the same likelihood. A precise estimate of additional learning may then come chiefly from the prior. Harder anchor items or an adaptive test can add information in this range (Montgomery and Cutler 2013); repeated easy items alone cannot resolve it.

Where the model is too weak to support a point estimate, bounds can be more informative. With deterministic threshold responses, no guessing, and externally calibrated item difficulties, each response pattern locates true knowledge in an interval $[L_{it}, U_{it}]$. Hence

$$\Delta X_i \in [L_{i2} - U_{i1}, U_{i2} - L_{i1}].$$

An independently justified no-forgetting assumption intersects this interval with $[0, \infty)$. A perfect score supplies a lower bound, with an upper bound only if one is known externally. With stochastic responses or guessing, these become probabilistic statements requiring an explicit response model, not deterministic bounds. Reporting wide uncertainty for a ceiling respondent is preferable to assigning that person either zero learning or a precise gain determined mainly by assumptions.

D Questionnaire coverage and validation details

D.1 Bounded illustration

For Figure 1, let initial knowledge be uniform on $[0, M]$. A nonempty region $[l, u]$ has probability $(u - l)/M$, conditional mean $(l + u)/2$, and conditional variance $(u - l)^2/12$. Weighting these moments by the region probabilities recovers the unconditional moments. For a general density f , the conditional mean in a region is $\int_l^u xf(x) dx / \Pr(l \leq X \leq u)$; the integral alone is its contribution to the unconditional mean. The same distribution determines the ceiling probabilities and the knowledge moments.

D.2 Controlled questionnaire comparisons

Our controlled comparison holds each simulated population and its realized learning fixed while changing the questionnaire. Each world supplies a bank of 30 items. Nested subsets contain 5, 10, 15, or 30 questions. Easy difficulties occupy the .05–.60 quantiles of the population’s gamma distribution; broader difficulties occupy the .05–.95 quantiles. The same item quantile ranks and response uniforms are used across designs. Discrimination is 3 and the probability of a correct guess is .25. We use 100 replications of 1,000 respondents for each learning form, without the poll-resemblance filter. Initial knowledge follows a gamma distribution with shape and rate one. The independent learning rate is lognormal with log mean $\log(.5)$ and log standard deviation .5. True gain is $X_{i1}R_i$ under proportional learning, R_i under additive learning, and $R_i \min(1/X_{i1}, 5)$ under catch-up learning. An item of difficulty d is answered correctly with probability $.25 + .75 \logit^{-1}[3(\log X - \log d)]$. Response draws are independent across waves and shared across questionnaire designs within a replication.

D.3 Domain coverage

The two-domain simulation fixes total gain at the person’s mean initial domain knowledge times a lognormal rate. Both initial domain traits are independent gamma variables with shape one; the rate has log mean $\log(.5)$ and log standard deviation $.5$. Domain weights are one half. Random allocation assigns all new knowledge to one domain, adding twice the target total gain there. The paired questionnaires share all realized knowledge, item-bank draws and response draws. A one-domain questionnaire uses ten first-domain items; the balanced questionnaire uses the first five from each domain. Every item has discrimination 3 and correct-guess probability $.25$.

Table D1: Median correlation with equally weighted total true gain across 100 replications. Each questionnaire has ten questions. Exposure identifies the domain receiving all new information; results for all respondents pool the two randomized exposure groups.

Questionnaire	Exposure	Initial	Gain	Follow-up
Both domains	All	0.50	0.04	0.53
Both domains	First domain	0.50	0.03	0.53
Both domains	Second domain	0.49	0.04	0.53
First domain	All	0.39	0.03	0.42
First domain	First domain	0.39	0.06	0.56
First domain	Second domain	0.39	0.00	0.39

D.4 Disjoint-item validation

We use the reviewed historical participant cohorts and their selected pre/post occasions at the same upstream revision as the aggregate analysis. Participants need scored responses to every matched item at both occasions; missing questionnaires and unresolved responses remain missing. Five-fold training/test splits keep discussion groups together. People without a group identifier form separate fold units. Random item partitions are compared with partitions choosing the easiest items according to baseline correctness in the training sample. Baseline residualization is also fitted only in that training sample. Neither held-out responses nor validation items enter either fitted operation. Within a test fold, every method uses the same participants and target items.

The procedure repeats the splits 100 times. We first take the median correlation, or paired correlation difference, across usable folds within each split, then report its median and 5th–95th percentile across splits. These intervals describe variation with the respondent and item partitions; they are not confidence intervals for a population parameter. Constant scores and constant validation gains yield undefined correlations, whose counts remain in the outputs. The sample retains 5,813 participants across 21 eligible polls; 17 participants are excluded for incomplete paired item responses. The item-level cohort and complete-case rule differ from those of the aggregate-score regressions; the samples are not interchangeable.

The empirical item validation retains the historical selected occasions. In the original poll-specific designs these include both pre-arrival-to-exit and other baseline-to-follow-up comparisons; selected t_1 and t_2 labels should not be read as a uniform arrival-to-exit intervention. Complete-case exclusions apply to the whole matched battery before either scoring rule is compared.

Table D2: Historical participant sample available to the disjoint-item comparison. Complete participants have scored responses for every matched item at both selected occasions. Excluded counts reflect incomplete paired-item responses, including unavailable forms; these responses are not assigned zero credit.

Poll	Available	Complete	Excluded	Items
Australia Republic (1999)	347	347	0	12
BTP General Election (2004)	248	248	0	9
BTP Health Education (2005)	454	454	0	6
BTP National (2003)	244	244	0	11
BTP Presidential Primaries (2004)	217	217	0	7
Bulgaria Crime (2002)	278	278	0	7
CPL (1996)	216	216	0	7
Europolis (2009)	348	348	0	6
New Haven (2004)	131	127	4	8
NIC (1996)	456	456	0	11
NIC2 (2003)	340	340	0	11
San Mateo (2008)	238	236	2	8
SWEPCO (1996)	232	232	0	5
Tomorrow's Europe (2007)	332	330	2	11
UK Crime (1994)	299	299	0	7
UK EU (1995)	224	224	0	5
UK General Election (1997)	275	275	0	15
UK Health (1998)	228	228	0	6
UK Monarchy (1996)	258	258	0	9
WTU (1996)	230	230	0	5
Zeguo (2005)	235	226	9	4